

```
In [1]: # PROJECT: CUSTOMER PERCEPTION TOWARDS RESTAURANTS IN NEPAL
```

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: sns.set_theme(style="whitegrid")
plt.rcParams.update({'font.sans-serif': 'Arial', 'font.family': 'sans-serif'})
```

```
In [ ]: #Load Dataset
```

```
In [6]: file_path = "C:/Users/admin/Downloads/Customer Survey.xlsx"
```

```
In [8]: df = pd.read_excel(file_path)
```

```
In [9]: df.columns = [col.strip() for col in df.columns]
```

```
In [ ]: #COLUMN RENAMING & PREPROCESSING
```

```
In [10]: col_mapping = {
    'Age group ': 'Age_Group',
    'Gender ': 'Gender',
    'Occupation ': 'Occupation',
    'Monthly Income': 'Monthly_Income',
    'How did you first learn about Korean restaurants?': 'Discovery_Source',
    'How often do you visit Korean restaurants?': 'Visit_Frequency',
    'What is the primary reason for choosing a Korean restaurant?': 'Primary_Reason',
    'How comfortable are you trying new Korean dishes?': 'Comfort_New_Dishes',
    'Which factor influences your restaurant choice the most?': 'Top_Influencing_Fa
}
```

```
In [12]: likert_cols = {
    df.columns[10]: 'Variety',
    df.columns[11]: 'Taste',
    df.columns[12]: 'Quality',
    df.columns[13]: 'Authenticity',
    df.columns[14]: 'Portion_Size',
    df.columns[15]: 'Affordability',
    df.columns[16]: 'Price_Fairness',
    df.columns[17]: 'Promotions',
    df.columns[18]: 'Staff_Courteous',
    df.columns[19]: 'Service_Prompt',
    df.columns[20]: 'Staff_Communication',
    df.columns[21]: 'Music_Ambiance',
    df.columns[22]: 'Decor',
    df.columns[23]: 'Cleanliness',
    df.columns[24]: 'KDrama_Influence',
    df.columns[25]: 'KPop_SocialMedia',
    df.columns[26]: 'Cultural_Motivation',
    df.columns[27]: 'Overall_Satisfaction',
    df.columns[28]: 'Revisit_Intention',
```

```
df.columns[29]: 'Recommendation'
}
```

```
In [13]: df = df.rename(columns={**col_mapping, **likert_cols})
```

```
In [14]: #LIKERT TEXT TO NUMERIC CONVERSION (1 TO 5 SCALE)
```

```
In [15]: likert_map = {
    'Strongly Disagree': 1,
    'Disagree': 2,
    'Neutral': 3,
    'Agree': 4,
    'Strongly Agree': 5
}
```

```
In [16]: for col in likert_cols.values():
    df[col] = df[col].map(likert_map)
```

```
In [17]: #COMPUTE CONSTRUCT COMPOSITE SCORES
```

```
In [20]: df['Food_Quality_Score'] = df[['Variety', 'Taste', 'Quality', 'Authenticity', 'Port
df['Pricing_Score'] = df[['Affordability', 'Price_Fairness', 'Promotions']].mean(ax
df['Service_Quality_Score'] = df[['Staff_Courteous', 'Service_Prompt', 'Staff_Comm
df['Ambiance_Cleanliness_Score'] = df[['Music_Ambiance', 'Decor', 'Cleanliness']].m
df['Pop_Culture_Influence_Score'] = df[['KDrama_Influence', 'KPop_SocialMedia', 'Cu
df['Customer_Loyalty_Score'] = df[['Overall_Satisfaction', 'Revisit_Intention', 'Re
```

```
In [21]: df['Age_Group'] = df['Age_Group'].astype(str).str.replace('â€‘', '-').str.replace('
```

```
In [22]: df.to_excel("Cleaned_Numeric_Survey_Data.xlsx", index=False)
print("Numeric data exported successfully to 'Cleaned_Numeric_Survey_Data.xlsx'.")
```

Numeric data exported successfully to 'Cleaned_Numeric_Survey_Data.xlsx'.

```
In [23]: #DESCRIPTIVE STATISTICS
```

```
In [24]: constructs = ['Food_Quality_Score', 'Pricing_Score', 'Service_Quality_Score',
    'Ambiance_Cleanliness_Score', 'Pop_Culture_Influence_Score', 'Custome

summary_stats = df[constructs].agg(['mean', 'std', 'min', 'max']).round(2).T
summary_stats.columns = ['Mean', 'Std_Dev', 'Min', 'Max']
print("\n=== CONSTRUCT SUMMARY STATISTICS (1-5 SCALE) ===")
print(summary_stats)
```

```
=== CONSTRUCT SUMMARY STATISTICS (1-5 SCALE) ===
```

	Mean	Std_Dev	Min	Max
Food_Quality_Score	4.10	0.95	1.0	5.0
Pricing_Score	3.78	1.03	1.0	5.0
Service_Quality_Score	3.97	1.05	1.0	5.0
Ambiance_Cleanliness_Score	4.01	1.04	1.0	5.0
Pop_Culture_Influence_Score	3.96	1.05	1.0	5.0
Customer_Loyalty_Score	4.08	0.99	1.0	5.0

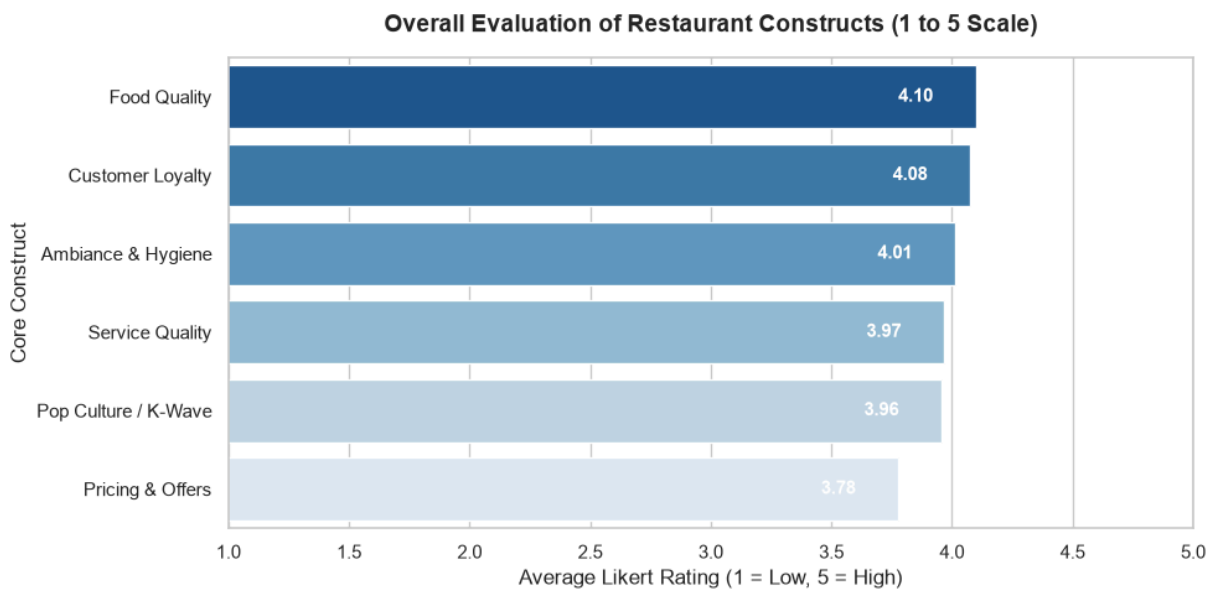
```
In [25]: #GRAPH 1: CORE CONSTRUCT MEANS COMPARISON BAR PLOT
```

```
In [27]: plt.figure(figsize=(10, 5))

construct_summary = pd.DataFrame({
    'Construct': ['Food Quality', 'Customer Loyalty', 'Service Quality',
                 'Ambiance & Hygiene', 'Pricing & Offers', 'Pop Culture / K-Wave']
    'Mean_Score': [
        df['Food_Quality_Score'].mean(),
        df['Customer_Loyalty_Score'].mean(),
        df['Service_Quality_Score'].mean(),
        df['Ambiance_Cleanliness_Score'].mean(),
        df['Pricing_Score'].mean(),
        df['Pop_Culture_Influence_Score'].mean()
    ]
}).sort_values(by='Mean_Score', ascending=False)

ax1 = sns.barplot(
    data=construct_summary,
    x='Mean_Score',
    y='Construct',
    hue='Construct',
    palette='Blues_r',
    legend=False
)
plt.title('Overall Evaluation of Restaurant Constructs (1 to 5 Scale)', fontsize=14)
plt.xlabel('Average Likert Rating (1 = Low, 5 = High)', fontsize=12)
plt.ylabel('Core Construct', fontsize=12)
plt.xlim(1, 5)
for p in ax1.patches:
    width = p.get_width()
    ax1.annotate(f'{width:.2f}',
                 (width - 0.25, p.get_y() + p.get_height() / 2.),
                 ha='center', va='center', color='white', fontweight='bold', fontsi

plt.tight_layout()
plt.savefig('construct_means_comparison.png', dpi=300)
plt.show()
```

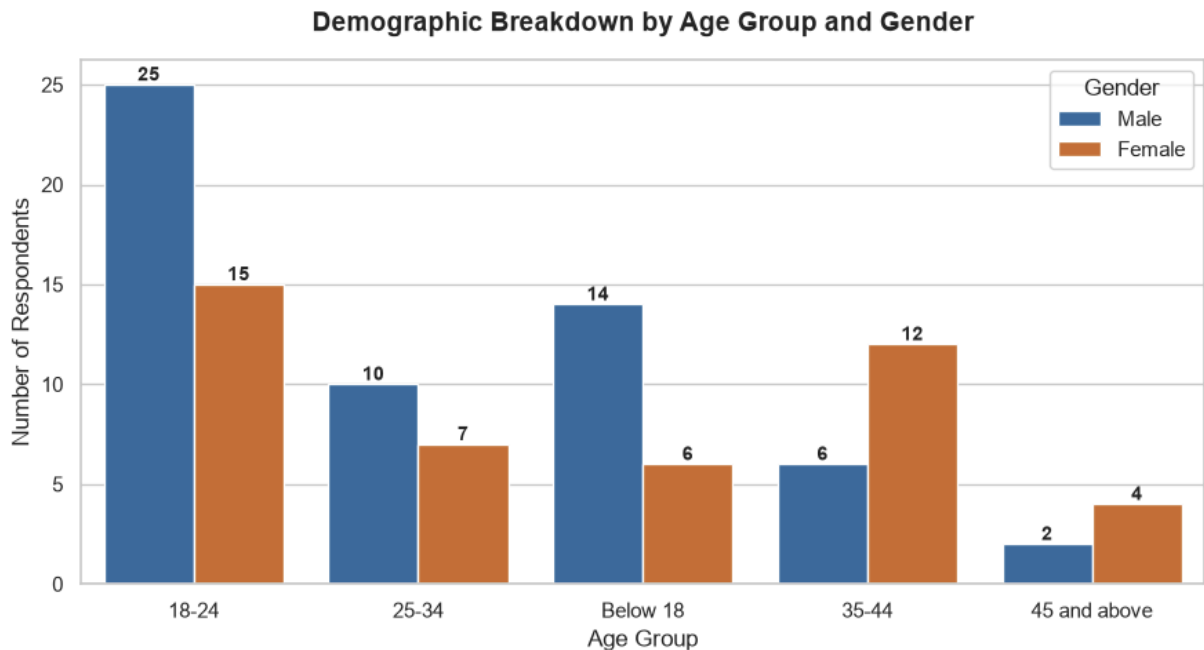


In [28]: *#GRAPH 2: DEMOGRAPHIC DISTRIBUTION (AGE & GENDER)*

```
In [31]: plt.figure(figsize=(9, 5))

# Using exactly 2 colors for the 2 gender categories in the dataset
ax2 = sns.countplot(data=df, x='Age_Group', hue='Gender', palette=['#2b6cb0', '#dd6b20'])
plt.title('Demographic Breakdown by Age Group and Gender', fontsize=14, fontweight='bold')
plt.xlabel('Age Group', fontsize=12)
plt.ylabel('Number of Respondents', fontsize=12)
plt.legend(title='Gender', loc='upper right')
for p in ax2.patches:
    height = p.get_height()
    if not np.isnan(height) and height > 0:
        ax2.annotate(f'{int(height)}',
                    (p.get_x() + p.get_width() / 2., height),
                    ha='center', va='bottom', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.savefig('demographics_distribution.png', dpi=300)
plt.show()
```



In [32]: *#GRAPH 3: POP CULTURE INFLUENCE VS CUSTOMER LOYALTY*

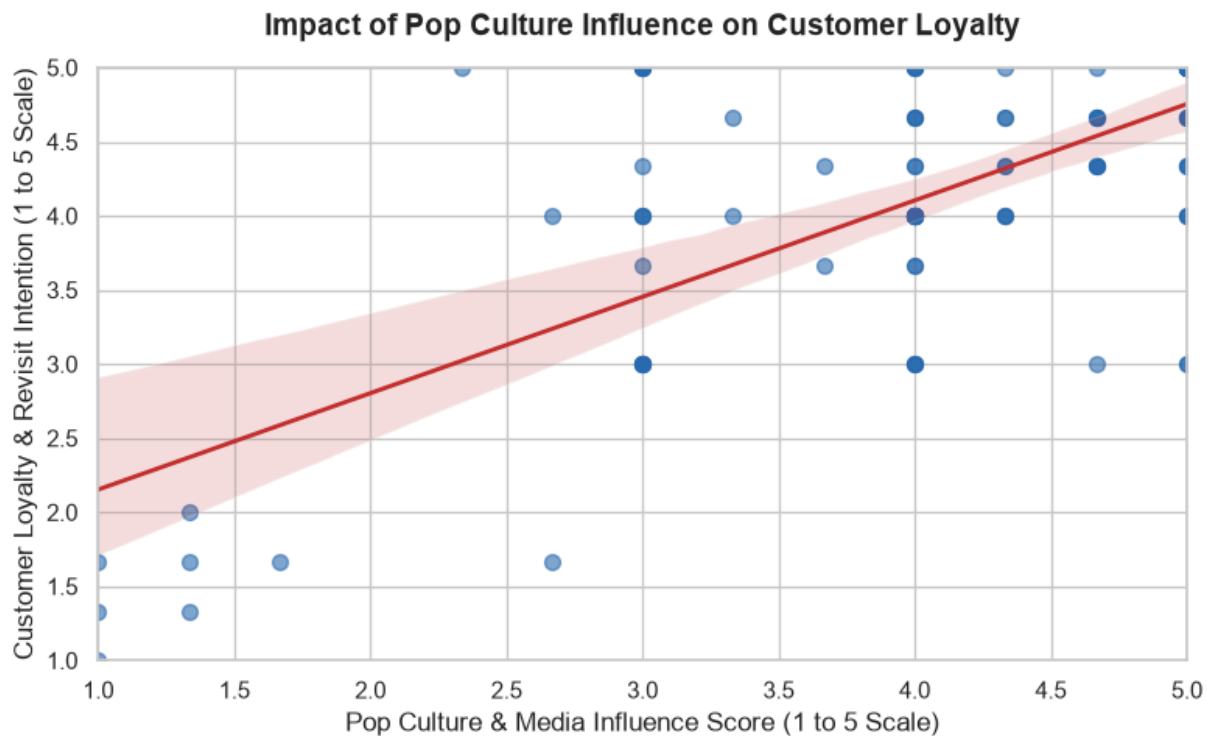
```
In [33]: plt.figure(figsize=(8, 5))

sns.regplot(
    data=df,
    x='Pop_Culture_Influence_Score',
    y='Customer_Loyalty_Score',
    scatter_kws={'alpha': 0.6, 'color': '#2b6cb0', 's': 50},
    line_kws={'color': '#c53030', 'linewidth': 2}
)

plt.title('Impact of Pop Culture Influence on Customer Loyalty', fontsize=14, fontw
```

```
plt.xlabel('Pop Culture & Media Influence Score (1 to 5 Scale)', fontsize=12)
plt.ylabel('Customer Loyalty & Revisit Intention (1 to 5 Scale)', fontsize=12)
plt.xlim(1, 5)
plt.ylim(1, 5)

plt.tight_layout()
plt.savefig('culture_vs_loyalty_regression.png', dpi=300)
plt.show()
```



In []: